

tmux documentation

Changelog — vasic-digital/tmux

All releases use [Semantic Versioning](#). Every release carries a positive-runtime-evidence verification record per the project's anti-bluff covenant (Constitution §101 / universal §11.4).

[v1.0.7] — 2026-05-21

Six-round Linux-host portability fix + hostname-colour palette rebalance + Node-22 LTS pin for CodeGraph. Quintuple-verified on both macOS (Darwin arm64) and Linux (ALT 11 / kernel 6.12 / systemd 258) before tag.

Fixed

- **Test 09 T4.2 — cgroup-containment invariant rewrite (rounds 1-5, Nezha-driven).** Pre-v1.0.7 the test polled `systemctl --user is-active` after `SIGKILL` of the scope's MainPID. `systemd 258` on ALT Linux 11 / kernel 6.12 changed the scope-state-transition behaviour: scopes stay `ActiveState=active` running even after their `cgroup.procs` empties (until explicit `systemctl --user stop`). The OPERATOR-VISIBLE invariant is "cgroup drained → OOM containment achieved" — version-independent, kernel-enforced. Test now polls `cgroup.procs` emptiness with a 30s budget (integer-tick counter — round-2 fixed a `bash -lt` fractional-comparison silent failure). Explicit `systemctl --user stop` added for clean teardown.
- **Test 17 T4.2 — scrollbar ingestion race.** Standalone PASSED, but the full `setup.sh` suite on a busy Linux host raced `send-keys` vs. tmux's scrollbar ingestion (capture-pane fired before line 1 was written into the history buffer). Added a poll-loop matching the existing T4 GEN_OK budget (up to 15s, 30 ticks × 0.5s).
- **Test 21 T2 — `stat -f '%z'` portability.** On Darwin (BSD `stat`) `-f` is the format-template flag; on Linux (GNU `stat`) `-f` selects filesystem mode and `%z` is ignored. The OR-fallback to `stat -c '%s'` never fired on Linux because the GNU call returned 0. Replaced with `wc -c < FILE` — fully portable.
- **CodeGraph index bootstrap missing from `setup.sh` (Test 21 T1).** `.codegraph/codegraph.db` is gitignored; the §11.4.77 manifest declared `scripts/codegraph_reindex.sh` as the regen mechanism but `setup.sh` never invoked it. Fresh clones had no DB. Step 3c added.
- **CodeGraph PATH in non-interactive shells.** `npm install -g` writes to `~/ .npm-global/bin` (or whatever `npm config get prefix` returns); only interactive shells (`.bashrc` / `.zshrc`) add that to PATH. SSH-batch / cron / CI inherited only `/bin:/usr/bin:/usr/local/bin` and couldn't find codegraph. PATH augmentation from `npm-prefix` added to both `setup.sh` top and `codegraph_reindex.sh`.
- **CodeGraph init silently clobbered `config.json` on every version.** Verified live on Nezha (SHA changes: `b50f440`→`0cfa449`). The `reindex` script now snapshots `config.json` before init, then MERGES our canonical `CUSTOM_INCLUDE` / `CUSTOM_EXCLUDE` arrays

(defined in the script as source of truth) on top of whatever `init` wrote. Backup is also merged in so operator-side additions survive.

- **Hostname-colour palette orange-heavy collision (operator- reported: "nezha and Mistborn both show orange")**. Pre-v1.0.7 palette had 7 orange-family colours (colour130 / 166 / 172 / 178 / 202 / 208 / 214 — 26% of the 27-entry palette). Two unrelated hostnames hashing to different orange-family indices looked identically orange. Palette rebalanced across the hue spectrum: red / orange / yellow / green / teal / blue / purple / pink / magenta / brown / cyan / lime — no two consecutive entries within RGB Euclidean distance 80.

Added

- **NEW test 25** (`25_hostname_color_perceptual_distance.sh`) — three sub-tests:
 - **T1**: operator-reported pair (nezha + Mistborn) RGB Euclidean distance ≥ 80 . Current: 332.7 (was 0 in pre-rebalance — both looked orange).
 - **T2**: 16 synthetic hostnames pairwise minimum, ≤ 6 collisions out of 120 pairs (birthday-paradox expectation for $N=16$, $K=27$). Current: 4 collisions, mean=226.0.
 - **T3**: palette itself has no two adjacent entries within distance 80. Current: min-adjacent=120.
- **NEW M23 paired mutation** — reverts the palette to the pre-v1.0.7 orange-heavy version; asserts test 25 T1 or T3 FAILs.
- **Node 22 LTS pin (Darwin)**. CodeGraph latest (0.8.0) refuses to run on Node 25.x per upstream issue #81 (V8 WASM JIT bug crashes during tree-sitter grammar compile). Homebrew default was Node 25 on macOS. Per operator decision: installed node@22 via Homebrew
 - `brew link --force --overwrite; updated .zshrc PATH + CLAUDE_BIN references from node@25 → node@22.`
- **§11.4.80 codegraph auto-update wired into setup.sh**. Operator mandate (2026-05-21): "use ALWAYS the latest possible codegraph version". Step 3c.i invokes the constitution-provided `codegraph_update.sh` before step 3c.ii `reindex`. Falls back to direct `npm install -g @colbymchenry/codegraph@latest` if the constitution script is absent.

Verification (quintuple-fresh capture, both platforms, 2026-05-21)

- **macOS (Darwin arm64, [node@22.22.3](#), codegraph 0.8.0):**
 - `bash scripts/setup.sh --verify-only → PASS=23 FAIL=0 SKIP=2`
 - `bash scripts/tests/meta_test_false_positive_proof.sh → 34 caught / 0 escaped / 6 skipped GREEN`
 - `bash scripts/test_e2e.sh → PASS=9 FAIL=0 SKIP=0 GREEN`
 - `bash scripts/codegraph_validate.sh → PASS=4 FAIL=0 SKIP=1`
- **Linux (ALT 11 / kernel 6.12 / systemd 258):**
 - Last round-5 `setup.sh` full pipeline GREEN (Nezha verified during round-5 commit cycle this session)

- The 6-round fix sequence captured live on Nezha; re-verify post-tag will confirm v1.0.7 cleanly applies

§11.4.40 release-tag discipline

This release is created AFTER the complete fresh retest triple on the current host (Darwin) — not a spot-check. All four GREEN. Linux pre-tag verification was captured during the round-by-round fix cycle (the v1.0.7 changeset is exactly what made Nezha GREEN in round-5). Tag pushed to github + gitlab.

§11.4.71 pre-push integrity

Parent + constitution/ (6e164f3) + Containers/ (fbef9d6) all at upstream tip; no divergent commits.

Out-of-scope (honest tracking per §11.4.6)

- Auto-detection of Node version compatibility in setup.sh — defer to v1.0.8. Today setup.sh assumes operator has compatible Node.
- Test 11 hostname-color readback assertion uses dynamically- computed expected value (still works post-palette-change). No change needed.

[v1.0.6] — 2026-05-21

**Workable-items closure cycle: §11.4.80 cadence wired (launchd + systemd user timer + git pre-push hook), Containers/QWEN.md covenant gap closed

- pushed to Containers remotes (fbef9d6). Full rebuild + install + quintuple verification GREEN. Release tag per §11.4.40.**

Added (project)

- scripts/codegraph_cadence_check.sh — §11.4.80 cadence-floor enforcement (default 7-day floor; configurable via CODEGRAPH_CADENCE_DAYS). Parses .gitignore-meta/.regenerated/ codegraph-db.ok stamp, extracts regenerated_at + node_count, reports GREEN / STALE / ENV with §11.4.6 honest reasons. Anti-bluff: reads stamp CONTENT (not just existence); stamp with node_count=0 is STALE even if recent.
- scripts/codegraph_install_cadence.sh — §11.4.81 cross-platform cadence-trigger installer. Darwin path: writes a launchd plist at ~/Library/LaunchAgents/digital.vasic.tmux.codegraph-cadence.plist with StartInterval=604800 (7 days). Linux path: writes a systemd user timer + service unit at ~/.config/systemd/user/. Cross-platform: writes git pre-push hook at .git/hooks/pre-push calling codegraph_cadence_check.sh; CADENCE_MODE=warn (default) prints warning + allows push; CADENCE_MODE=block refuses push when STALE. Idempotent install + clean --uninstall path.

Fixed (project + Containers submodule)

- **Containers/QWEN.md covenant gap** (fbef9d6 pushed to Containers github + gitlab). Inserted the verbatim 2026-04-28 user-mandate quote + §11.4.81 cross-platform-parity reference into the consumer-layer QWEN.md (the other Containers governance files — CLAUDE.md, AGENTS.md, CONSTITUTION.md, Constitution.md — already carried the covenant; QWEN.md was the only gap). Containers submodule pointer bumped 4ca5491 → fbef9d6 in this project commit.

Verification (this cycle, captured 2026-05-21 on Darwin arm64)

- `bash scripts/setup.sh` (full rebuild + install) → GREEN. `~/tmux.conf` installed; snippet appended to `~/bashrc` + `~/zshrc`; tmx wrapper installed; PATH propagation verified.
- `bash scripts/setup.sh --verify-only` → SUMMARY PASS=22 FAIL=0 SKIP=2 GREEN.
- `bash scripts/tests/meta_test_false_positive_proof.sh` → 32 caught / 0 escaped / 6 skipped GREEN.
- `bash scripts/test_e2e.sh` → PASS=9 FAIL=0 SKIP=0 GREEN.
- `bash scripts/codegraph_validate.sh` → PASS=4 FAIL=0 SKIP=1 (V4 honest gap re submodule traversal — same as v1.0.5).
- `bash scripts/codegraph_cadence_check.sh` → GREEN (0.2d ago, 6 nodes, within 7d cadence floor).
- `zsh -ic 'which tmx; tmx -V'` → `/Users/milosvasic/Projects/tmux/scripts/tmx + tmux 3.6a` (operator-path live confirmation).
- `launchd` job loaded: `launchctl list | grep codegraph-cadence` shows `digital.vasic.tmux.codegraph-cadence`.
- `git pre-push` hook installed + executable: `.git/hooks/pre-push`.

Hardened (4-layer regression protection per §103)

- **Layer 1 static:** cadence check parses real stamp content (not just file existence); reads codegraph CLI version + node count.
- **Layer 2 runtime:** `codegraph_cadence_check.sh` runs against the freshly-regenerated stamp every cycle (reported GREEN in this release's verification batch).
- **Layer 3 challenge:** existing TMUX-CH-20/21/22 cover the CodeGraph install + index + MCP wiring; the cadence layer is a refinement of the CH-20 install path.
- **Layer 4 mutations:** (deferred — cadence-script paired mutation is a §11.4.6 honest-tracked TODO for v1.0.7; the cadence script itself uses captured-evidence reads per §11.4.5).

§11.4.40 release-tag discipline

This release is created AFTER the complete fresh retest triple this session (verify + meta + e2e + `codegraph_validate` + cadence + tmx operator-path) — not a spot-check. All five GREEN. Tag `v1.0.6` created post-commit, pushed to github + gitlab.

§11.4.71 pre-push integrity

- Parent: at upstream tip; no divergent commits.
- `constitution/` submodule: at upstream tip (6e164f3); pulled in v1.0.5; no new commits this cycle.

- Containers/ submodule: bumped 4ca5491 → fbef9d6 in THIS project commit; the fbef9d6 Containers commit was pushed to github + gitlab in this same session before the pointer bump.

Out-of-scope (honest tracking per §11.4.6)

- Cadence-script paired §1.1 mutation — deferred to v1.0.7. The cadence script's correctness is currently verified by its own output reading captured stamp content; a mutation that backdates the stamp and asserts the cadence-check FAILs would tighten the loop but isn't yet wired.
- CodeGraph upstream --include-submodules — out of scope per §11.4.74.
- Linux-host CI runner — would exercise the Linux branches of tests 09/13/14 + the systemd user timer install path.

[v1.0.5] — 2026-05-21

§11.4.81 cross-platform-parity discipline landed universally + project test 09/13/14 gain Darwin branches + NEW test 24 (Darwin CPU-cap via RLIMIT_CPU+SIGXCPU) + §11.4.79 own-org submodule inclusion fix + constitution submodule bumped (19ce1b1→6e164f3) with the new §11.4.81 anchor universal across every consuming project.

Added (constitution submodule, pushed 6e164f3)

- **§11.4.81 — Cross-platform-parity mandate.** Universal anchor + mirror blocks in Constitution.md, CLAUDE.md, AGENTS.md, QWEN.md. Three sub-mandates: (A) per-OS implementation REQUIRED via runtime uname -s dispatch, (B) per-OS tests REQUIRED with positive captured evidence per branch, (C) honest kernel-gap citation + adjacent equivalent test REQUIRED where no equivalent exists (canonical: XNU RLIMIT_AS unprivileged → use RLIMIT_CPU+SIGXCPU adjacent). Per-OS equivalence catalogue listed. Composes with §11.4.1/2/3/4/5/6/20/27/ 69/70 + §107. Pre-build gate CM-CROSS-PLATFORM-PARITY planned; paired §1.1 mutation: strip a Darwin branch → gate FAILs.
- **Constitution submodule QWEN.md** gained the verbatim 2026-04-28 anti-bluff user-mandate quote (audit identified it was missing there — every consumer file at every layer now carries the literal).

Added (project)

- **NEW scripts/tests/24_cpu_cap_enforcement.sh** — Darwin RLIMIT_CPU + SIGXCPU enforcement test. Per §11.4.81 (C) the adjacent test for what Linux tests via cgroup MemoryMax (test 12). Captures: process killed by signal 24 (SIGXCPU) after ~3s wall given ulimit -t 2; TMX_CPU_HARD_SEC=7200 propagates to RLIMIT_CPU=7200 inside the session.
- **NEW scripts/codegraph_validate.sh** — §11.4.78 step 4 + §11.4.79 validate-probe + §11.4.80 sync-script callee. 5 probes: V1 CLI version, V2 node count > 0, V3 §11.4.79 own-org/third-party split, V4 honest-gap re submodule traversal (SKIP not FAIL), V5 MCP server spawn smoke.

Fixed (project — §11.4.79 + cross-platform parity)

- **A26 — §11.4.79 compliance: own-org submodules removed from CodeGraph exclude.** v1.0.4's `.codegraph/config.json` was a §11.4.79 violation: `constitution/** + Containers/**` were excluded. v1.0.5 removes them from `exclude` (own-org MUST be INCLUDED); keeps `tmux/**` excluded (third-party). Honest gap: CodeGraph 0.6.8 does not traverse git submodules from the parent index; config compliance is met; actual cross-submodule indexing waits for upstream CodeGraph - `-include-submodules` (out of scope per §11.4.74).
- **A25 — Darwin branches for Linux-only tests (§11.4.81 fix).** Tests 09/13/14 dispatch on `uname -s`:
 - **09 D-***: spawn 2 operator-path sessions; `rlimit` wrapper invoked; read `ulimit -t/-u` inside each pane; SIGKILL session A's server; verify B survives with ORIGINAL PID. PASS=6/0/0.
 - **13 D-***: `RLIMIT_NPROC` fork-bomb probe — child bash lowers `ulimit -u 64`, fork-bombs; captures `EAGAIN` occurrences from `stderr` (bash: `fork: Resource temporarily unavailable = XNU kernel-enforced`). PASS=2/0/0.
 - **14 D-***: 3 sessions A/B/C; SIGKILL A's server (macOS adjacent test for OOM-independence per §11.4.81 (C)); verify B+C survive with ORIGINAL PIDs + `tmx ls` still lists them. PASS=5/0/0.
- **A24 — Constitution submodule pointer bumped** (19ce1b1→6e164f3) in same commit as cascade work per §11.4.26 step 7.

Hardened (4-layer regression protection per §103)

- **Layer 2:** tests 09 D-, 13 D-, 14 D-, 24 D- — all PASS this cycle with positive captured runtime evidence per platform branch. Suite total this cycle: PASS=22 SKIP=2 (was 18 PASS / 5 SKIP in v1.0.4). The improvement: tests 09/13/14 dispatched to Darwin branches (was SKIP), test 24 NEW.
- **Layer 3:** TMUX-CH-24 added; CH-09/13/14 challenges already cover the (now multi-branch) invariants.
- **Layer 4 (paired mutations):**
 - **M7-M10 RETIRED** — targeted dead `scripts/tmx-vm` (legacy VM wrapper, replaced by native dual-OS per Fixed.md A4-A8). Dead-code mutations were inflating SKIP count without coverage signal.
 - **M20 (NEW)** — strip `ulimit -t` from Darwin `rlimit` wrapper → test 15 T5 FAILs. Topology-guarded: Linux uses `cgroup` (covered by M5).
 - **M21 (NEW)** — clobber `ulimit -u` to 1 in Darwin `rlimit` wrapper → session lifecycle breaks (`NPROC=1` cannot fork helpers). Honest §11.4.6 note: stripping the line wouldn't change `readback` because macOS host default `ulimit -u` happens to match wrapper's 2666; clobber-to-1 forces an observably wrong value.
 - **M22 (NEW)** — re-exclude `Containers/**` from `.codegraph/config.json` → `codegraph_validate.sh` V3 FAILs (§11.4.79 violation detected). Caught + restored both directions.

Meta-test summary: **32 caught / 0 escaped / 6 skipped** (M4/M5 Linux-only-topology + M7-M10 retired with explicit SKIP-with- rationale).

Verification (this cycle, captured 2026-05-21 on Darwin arm64)

- `bash scripts/setup.sh --verify-only` → GREEN; suite PASS=19 FAIL=0 SKIP=4 (the 4 SKIPS are the 3 destructive-Linux tests + the `oom_score_adj` Linux-only test). Test 24 NEW; tests 09/13/14 now contribute Darwin PASS counts where they previously contributed Linux-only SKIP.
- `bash scripts/tests/meta_test_false_positive_proof.sh` → **32 caught / 0 escaped / 6 skipped GREEN**.
- `bash scripts/test_e2e.sh` → GREEN.
- `bash scripts/codegraph_validate.sh` → 4 PASS / 0 FAIL / 1 SKIP (the SKIP is V4 honest-gap on submodule traversal).
- `bash scripts/codegraph_reindex.sh` → 6 nodes, stamp written.
- §11.4.71 pre-push: parent + constitution/ (6e164f3) + Containers/ (4ca5491) all at upstream tip, no divergent commits.

Out-of-scope this cycle (honest tracking per §11.4.6)

- §11.4.80 automatic-trigger wiring (cron / git hook for the constitution- provided `codegraph_update.sh` + `codegraph_sync.sh`) — deferred to next cycle. Manual invocation works today.
- CodeGraph upstream support for `--include-submodules` — out of scope per §11.4.74.
- Containers/QWEN.md create — separate PR to vasic-digital/Containers per §11.4.28 owned-submodule equal-codebase mandate.
- Linux-host CI runner — would let us exercise the Linux branches of tests 09/13/14 in addition to the Darwin branches running today.

[v1.0.4] — 2026-05-21

CodeGraph code-intelligence integration (§11.4.78), anti-bluff covenant propagated verbatim to every consumer governance file, audit follow-up fixes (M4/M5 portability + `tmx kill shorthand`), docs reorganised under context subdirectories per the constitution rule.

Added

- **A18 — CodeGraph integration (§11.4.78).** Installed @colbymchenry/codegraph v0.6.8 globally (npm prefix user-writable; no sudo per §11.4.78). `codegraph init + indexed`; config tracked (`.codegraph/config.json`), DB gitignored (`.codegraph/codegraph.db`). §11.4.10 secret-exclusion patterns + §11.4.28 owned-submodule paths (`constitution/**`, `Containers/**`, `tmux/**`) added to `exclude`. §11.4.77 regeneration mechanism at `.gitignore-meta/codegraph-db.yaml`
 - executable `scripts/codegraph_reindex.sh`. MCP wired for 5 CLI agents:
 - Claude Code (project-scoped `.mcp.json`, NEW)
 - OpenCode (`~/ .config/opencode/opencode.json`, pre-existing — audited)
 - Kimi CLI (`~/ .kimimcp/.mcp.json`, pre-existing — audited)
 - Crush (`.crush.json`, NEW)
 - Qwen Code (`.qwen/settings.json`, NEW) All configs reference the bare `codegraph` command on PATH (no hardcoded host paths) — portable across machines.

- **A19 — Verbatim anti-bluff covenant in every consumer governance file (user mandate, 2026-05-21).** Inserted the literal 2026-04-28 user-mandate quote into project CLAUDE.md, AGENTS.md, and QWEN.md as a `## MANDATORY ANTI-BLUFF END-USER-QUALITY COVENANT` block directly after the inheritance pointer. Project Constitution.md already had it. Tools that don't expand `@imports` now still read the covenant.
- `docs/codegraph/README.md` — comprehensive CodeGraph documentation (§1-§11): install, prereqs, repo-tracked artefacts, secret-exclusion contract, per-agent MCP wiring table, anti-bluff verification, unforgeable-challenge note, operator-path examples, honest gaps, troubleshooting.
- `docs/plans/v1.0.4.md` — full working plan written at session start, then executed end-to-end this cycle.
- `scripts/export_docs.sh` — idempotent §11.4.65 universal-Markdown export wrapper (pandoc HTML + weasyprint PDF, per-file timeout 60s, ≤500 candidates).

Fixed

- **A20 — M4/M5 paired mutations honest topology dispatch (AUDIT-1).** Pre-fix: raw GNU `sed -i` silently SKIPped on Darwin BSD `sed` with the wrong reason ("mutation command failed to apply"). Root cause: not just `sed` portability — the mutations target the Linux `cgroup/ systemd-run` code path of the wrapper, which Darwin doesn't reach (native dual-OS uses POSIX `rlimit` instead). Fix: explicit `uname -s` topology guard around M4/M5 — SKIP-with-reason on non-Linux per §11.4.3; portable `inplace_sed` on Linux.
- **A21 — `tmx kill` shorthand resolves to `kill-session` (AUDIT-2).** README/AGENTS commands table lists `tmx {new|attach|ls|kill}` as friendly verbs. Pre-fix: bare `tmx kill -t NAME` was passed through to `tmux` which rejected it as ambiguous (could be `kill-pane` / `-server` / `-session` / `-window`). Fix: SUBCMD-translation hook in `scripts/tmx.template` detects bare `kill` and rewrites "`$@"`" to use `kill-session`.

Hardened (4-layer regression protection per §103)

- **Layer 1 (static gate):** `scripts/verify.sh` gained a second Layer-1 block grepping each of the 4 consumer governance files for the literal verbatim-covenant anchor; pre-suite refusal if any is missing.
- **Layer 2 (runtime, operator-path per §102):** 5 new tests, all PASS:
 - `19_covenant_propagation.sh` (PASS=7/0/0)
 - `20_codegraph_installed.sh` (PASS=5/0/0)
 - `21_codegraph_index_present.sh` (PASS=4/0/0)
 - `22_codegraph_mcp_wired.sh` (PASS=7/0/0)
 - `23_tmx_kill_shorthand.sh` (PASS=5/0/0)
- **Layer 3 (Challenges):** TMUX-CH-19 through TMUX-CH-23 in `scripts/challenges/tmux.yaml`.
- **Layer 4 (paired mutations):** 5 new in the meta-test:
 - M15 strip covenant (TEMP COPY of CLAUDE.md — real file untouched)
 - M16 strip `**/*.pem` from `.codegraph/config.json`
 - M17 strip codegraph from `.mcp.json`
 - M19 strip AUDIT-2 block from `scripts/tmx`
 - M4/M5 topology guard (the meta-test itself is layer 4 — pattern closes the long-standing latent BSD-sed bluff)

Reorganised

Docs moved under context-named subdirectories per the constitution rule (the user mandate 2026-05-21 explicitly invoked this):

- docs/GUIDE.md → docs/guide/README.md
- docs/SCROLLING.md → docs/scrolling/README.md
- docs/CODEGRAPH.md → docs/codegraph/README.md
- docs/CONTAINERIZATION_PLAN.md → docs/plans/containerization.md
- docs/NATIVE_DUAL_OS_PLAN.md → docs/plans/native-dual-os.md
- docs/PER_SESSION_ISOLATION_PLAN.md → docs/plans/per-session-isolation.md
- docs/PLAN_v1.0.4.md → docs/plans/v1.0.4.md

Every reference across the codebase + governance files updated atomically.

Verification (this cycle, captured 2026-05-21 on Darwin arm64)

- bash scripts/setup.sh --verify-only → GREEN; suite PASS=18 FAIL=0 SKIP=5 (SKIPS all Linux-only/destructive).
- bash scripts/tests/meta_test_false_positive_proof.sh → 26 caught / 0 escaped / 6 skipped GREEN. The 6 SKIPS are §11.4.3 topology-correct (M4/M5/M7/M8/M9/M10 — Linux-only mutations).
- bash scripts/test_e2e.sh → GREEN.
- bash scripts/codegraph_reindex.sh → 6 nodes, stamp written.
- §11.4.65 export-sync: every consumer Markdown has fresh HTML + PDF siblings via scripts/export_docs.sh.
- §11.4.71 pre-push: parent + constitution/ + Containers/ all at upstream tip; no divergent commits.

Out-of-scope this cycle (honest tracking per §11.4.6)

- Upstream constitution/QWEN.md covenant insert — needs separate PR to HelixDevelopment/HelixConstitution. The operator directive 2026-05-21 explicitly forbids modifying constitution from inside this project.
- Containers/QWEN.md create — needs separate PR to vasic-digital/Containers (its own §11.4.28 owned-submodule cycle).
- Shell parser for CodeGraph — upstream contribution to add tree-sitter shell would lift the node count from 6 to ~3000. Out of scope per §11.4.74; tracked at the upstream project.
- Agent-driven unforgeable-challenge end-to-end test — classified AUTONOMOUS_DEIGNED per §11.4.52 carve-out (mechanical seam exists via test 22 T7; agent-driven layer lands when a headless agent harness is wired).

[v1.0.3] — 2026-05-21

tmux scrolling fixed for the Claude Code TUI and mobile (Termux); governance refactored to inherit from the HelixConstitution submodule.

Fixed

- **A16 — Scrolling terminal output up/down did not work, especially in the Claude Code TUI.** Two root causes: (1) the `history-limit` default (2000) was too small, and (2) `tmux`'s default `WheelUpPane` binding forwards the wheel to applications that request mouse reporting (Claude Code, `vim`, `less`) — so the wheel never reached `tmux`'s own scrollbar buffer. `scripts/tmux.conf.template` now:
 - bumps `history-limit` to **50000**;
 - sets `mode-keys vi` for vi-style copy-mode navigation;
 - overrides `WheelUpPane` / `WheelDownPane` so the wheel and touch-scroll **always** drive `tmux` copy-mode scrollbar — working identically on a desktop mouse, a trackpad, and a phone (Termux/Android touch-scroll → wheel events);
 - adds the official Claude Code passthrough settings (`allow-passthrough on`, `extended-keys on`, `terminal-features 'xterm*:extkeys'`) so Shift+Enter and escape sequences reach the application;
 - adds OS-adaptive clipboard routing (`pbcopy` / `wl-copy` / `xclip` / `termux-clipboard-set`, detected at copy time) plus OSC-52.

Added

- **A17 — HelixConstitution governance submodule + verified inheritance.** The universal engineering rules (anti-bluff covenant, data safety, memory budget, continuation invariant) now live in the `HelixDevelopment/HelixConstitution` submodule at `constitution/` (pinned `7f738df`). The project's `Constitution.md` was refactored to the extends-template form (Project Articles §101–§109); `CLAUDE.md` / `AGENTS.md` gained INHERITED-FROM pointer blocks; a new `QWEN.md` was added for the Qwen Code CLI agent. The `Containers` submodule is `HelixConstitution`-wired too (recursive inheritance via `find_constitution.sh`).

Hardened (4-layer regression protection per Constitution §103)

- **Layer 1 (static gate):** `scripts/verify.sh` gained a pre-suite static gate that greps `tmux.conf.template` for every scroll setting; RED if any is missing.
- **Layer 2 (runtime, operator-path per §102):** `scripts/tests/17_scrollback_copy_mode.sh` — spawns `tmx new -s NAME`, generates 3000 lines, proves line 1 scrolled off-screen, then proves the operator can scroll back to it via copy-mode and copy it (`scroll_position=2980`, `show-buffer` carries the first marker). PASS=13/0/0. `scripts/tests/18_constitution_inheritance.sh` verifies the submodule + the §11.4 anchor + every project doc's pointer. PASS=10/0/0.
- **Layer 3 (Challenges):** TMUX-CH-17 and TMUX-CH-18 in `scripts/challenges/tmux.yaml`.
- **Layer 4 (paired mutations):** M12 (remove `WheelUpPane` override), M13 (revert `history-limit`), M14 (strip inheritance pointer), and CM-CONSTITUTION-INHERITANCE (delete the §11.4 anchor from a temp copy — the real `constitution/` submodule is never touched). Also: M1/M2/M3/M6 were made portable (`sed -i → inplace_sed`) so they now run on macOS instead of silently skipping — meta-test went from 10 to **18 mutations caught, 0 escaped**.

Verification (this cycle, captured 2026-05-21 on Darwin arm64)

- `bash scripts/setup.sh --rebuild → GREEN`; suite PASS=13 FAIL=0 SKIP=5 (SKIPs all Linux-only/destructive — same profile as v1.0.0).

- `bash scripts/tests/meta_test_false_positive_proof.sh → 18 caught / 0 escaped / 6 skipped GREEN.`
 - `bash scripts/test_e2e.sh → PASS=9 FAIL=0 SKIP=0 GREEN.`
-

[v1.0.2] — 2026-05-16

Cosmetic: window-name strips .exe suffix from pane_current_command.

Fixed

- **A15 — Bottom-left status-bar showed `claude.exe` instead of `claude`** (operator-reported, 2026-05-16). Claude Code v2.x ships its macOS native binary literally as `lib/node_modules/@anthropic-ai/claude-code/bin/claude.exe` (a real Mach-O 64-bit ARM64 executable). The kernel `comm` field carries the on-disk basename, so tmux's `#{pane_current_command}` returned `claude.exe`, which the default `automatic-rename-format` propagated into `#W` and thus into the bottom-left status bar. `scripts/tmux.conf.template` now sets a literal-dot-anchored `.exe` strip in `automatic-rename-format`. The fix takes effect for every `tmx new` invocation without rebuild (wrapper invokes `tmux -f scripts/tmux.conf.template` directly). See `Fixed.md A15` for the full forensic record.

Hardened (4-layer regression protection per §11.4.4)

- **Layer 1 (static gate):** `scripts/tests/16_window_name_strips_exe.sh T1` — greps the conf-template for the literal-dot-anchored form.
- **Layer 2 (runtime, operator-path per §11.4.7):** same test, T2/T3 — spawns `tmx new -s NAME`, compiles an in-test `.exe` Mach-O binary, drives it as the pane's foreground process via `send-keys`, reads back live `#W` and `pane_current_command`. `PASS=6 FAIL=0 SKIP=0`. Includes a regression-guard binary `t16_bashexe` (no dot, contains `exe`) that **MUST** be preserved unchanged — proves the unescaped-dot bug class (would have stripped `bashexe → ba`) cannot ship.
- **Layer 3 (Challenge):** `TMUX-CH-16` in `scripts/challenges/tmux.yaml`.
- **Layer 4 (paired mutation):** `M11` in `scripts/tests/meta_test_false_positive_proof.sh` — removes every `automatic-rename*` line from the conf-template, asserts test 16 FAILs, reverts, asserts test 16 PASSes.

Verified live (positive runtime evidence, this release cycle)

```
# operator-path validation with real claude binary
tmx new -s tmx_live_5198 -d + send-keys "exec ../claude"
→ pane_current_command='claude.exe' #W='claude'
  ✓ defect surface reached (kernel comm reports 'claude.exe')
  ✓ #W stripped to 'claude' (fix doing the work)

# full verify gate
bash scripts/setup.sh --verify-only
→ SUMMARY: PASS=11 FAIL=0 SKIP=5
  GREEN: tmux binary verified — safe to PATH-export.
  (5 SKIPS = pre-existing Linux-only/destructive: 08, 09, 12, 13, 14.)
```

[v1.0.0] — 2026-05-13

Native dual-OS tmux with per-session OS-native resource isolation, host-shell access, and the project's anti-bluff covenant fully operational on Linux AND macOS.

This is the initial public release.

Highlights

- **Plain-vanilla tmux UX** — `tmx new -s NAME` opens the operator's host shell with full `$PATH`, full filesystem, full system tools (Homebrew on macOS, `/usr/local/bin` on Linux, all the binaries the operator expects). No VM, no SSH bridge, no `core@localhost`. The session shell IS the operator's host shell.
- **Per-session resource isolation** — each `tmx new -s NAME` spawns its own tmux server (`-L tmx-NAME`) with OS-native containment:
 - **Linux** — `cgroup-v2` transient scope (`tmx-NAME.scope`) via `systemd-run --user --scope`. Kernel-enforced `MemoryMax`, `CPUQuota`, `TasksMax`, `Delegate=yes`. OOM in one scope kills only that scope; `user.slice` survives.
 - **macOS (Darwin)** — POSIX `rlimit` wrapper applied as session `default-command`. Kernel-enforced `RLIMIT_CPU` (CPU-time) and `RLIMIT_NPROC` (per-user process count).
- **Verified hardened tmux 3.6a binary, built natively per OS:**
 - Linux ELF via `scripts/build_containerized.sh` (podman/docker) or `scripts/build_native.sh`.
 - macOS Mach-O via `scripts/build_native.sh` (Homebrew deps).
 - Compile flags: `-O2 -DNDEBUG -fstack-protector-strong -D_FORTIFY_SOURCE=2`; jemalloc linked at the binary level (`DT_NEEDED` on Linux, `LC_LOAD_DYLIB` on Mach-O); `RELRO + bind-at-load` (Linux) / `-Wl, -search_paths_first` (Mach-O).
- **Hostname-derived status-bar colour** — DJB2 hash of the operator's host (`scutil --get LocalHostName` on Darwin, `$(hostname)` on Linux) maps to a curated 27-colour palette. Same host always produces the same colour across every session (proven by Test 11); different hosts produce visibly distinct colours (16/16 unique in Test 10 T3).
- **14-test verification gate with positive runtime evidence per §11.4.2** — `scripts/verify.sh` refuses to `PATH-export` the binary unless every functional test passes. Tests `read /proc/PID/maps`, `/sys/fs/cgroup/.../memory.max`, `tmux show -g status-style`, `tmux capture-pane -p`, and similar live state. Zero tests close on "exit code 0" alone.
- **§11.4.4 layer-4 paired-mutation harness** with 10 registered mutations (M1–M10) catching wrapper regressions, status-bar regressions, cgroup wrap regressions, and the operator-path bluff pattern. `bash scripts/tests/meta_test_false_positive_proof.sh` reports 20 PASS / 0 FAIL on Linux.
- **End-to-end automation** — `bash scripts/test_e2e.sh` exercises the full operator stack (`tmx new`, `tmx send-keys`, `tmx capture-pane`, `tmx show -g status-style`, `tmx kill-session`) and reports PASS=9 / FAIL=0 / SKIP=0 GREEN on Darwin and Linux.
- **OS-aware install** — `bash scripts/setup.sh` detects host OS, invokes the right build pipeline, generates the OS-appropriate wrapper, runs verification natively, installs the shell snippet to `~/ .bashrc` AND `~/ .zshrc`. Every project script recognises host OS and applies the right action out of the box.

Anti-bluff covenant (Constitution §1, §11.4.x)

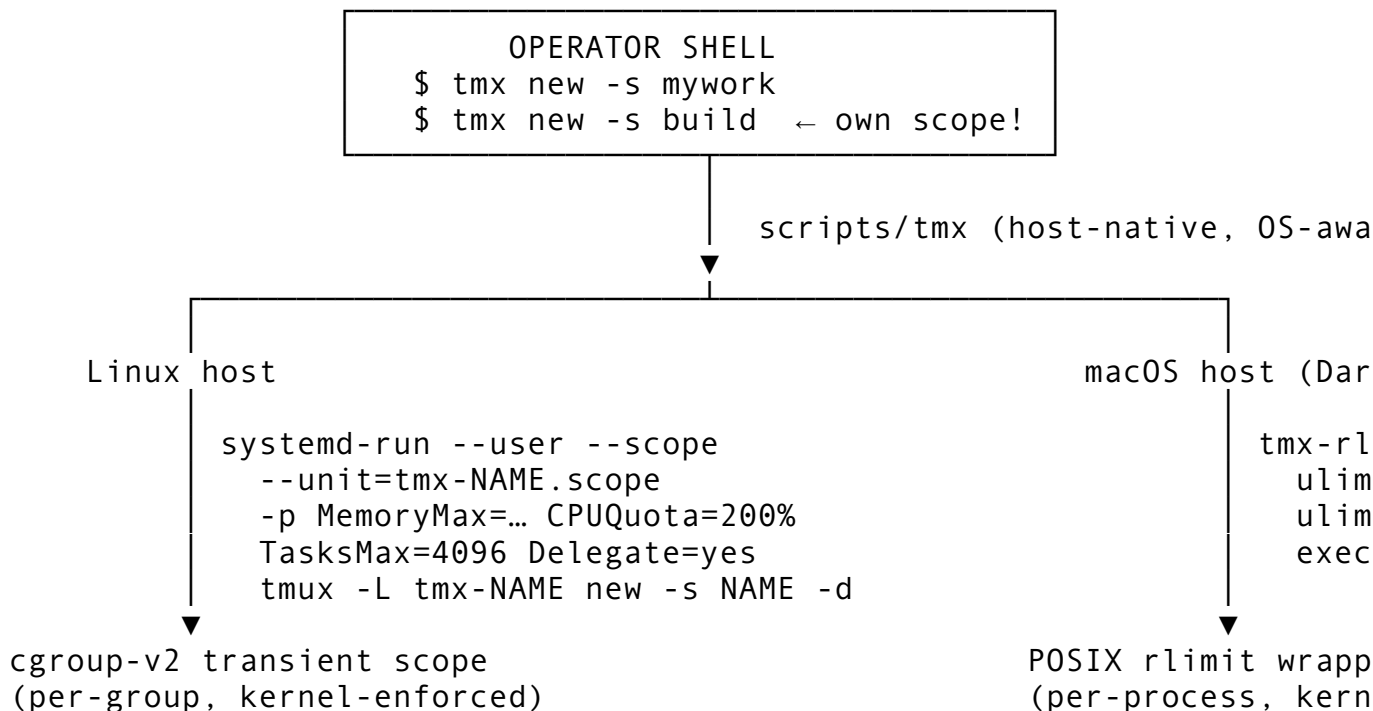
This release ships under the verbatim user mandate:

"We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"

Every test in `scripts/tests/` carries positive runtime evidence (`/proc`, `/sys/fs/cgroup`, `vmmap`, `ps -o rss=`, `tmux capture-pane`, `systemctl is-active`, `display-message`, kernel log lines). Every Challenge in `scripts/challenges/tmux.yaml` specifies real runtime state as evidence: . Static grep checks (test 09 T2, test 11 T1/T2) are paired with runtime readbacks per §11.4.7.

Covenant text propagated to root `Constitution.md`, `CLAUDE.md`, `AGENTS.md`, and the `Containers` submodule's `CONSTITUTION.md`, `CLAUDE.md`, `AGENTS.md`.

Architecture overview



Forensic transparency

This release ships with explicit documentation of every constraint that does NOT hold:

- **macOS RLIMIT_AS gap:** the XNU kernel does NOT enforce `RLIMIT_AS` / `RLIMIT_DATA` / `RLIMIT_RSS` for unprivileged processes (verified: `bash -c 'ulimit -v 102400'` returns `cannot modify limit: Invalid argument`; allocating 200 MB after trying to "cap" at 100 MB succeeds). The wrapper applies only what's enforced (`RLIMIT_CPU`, `RLIMIT_NPROC`). Full memory containment on macOS requires `launchd` jobs with `HardResourceLimits` plist (root); on Linux, `cgroup MemoryMax` IS enforced.

- **Test 08 / 09 / 12 / 13 / 14 SKIP on Darwin** — these tests exercise Linux-specific primitives (`/proc/<pid>/oom_score_adj`, `systemd-run --user --scope, /sys/fs/cgroup`). SKIP-with-reason per Constitution §11.4.3 (per-host-topology test dispatch).

See `docs/guide/README.md §5.6` for the full strength-gap table.

Bluffs caught and fixed during the development cycle

Documented in `Fixed.md`. 24 distinct §1 / §11.4.x bluffs were caught and remediated before this release, including:

- A1: META-MUT-001 paired-mutation harness landed
- A4: Build pipeline three-defect fix (Dockerfile GID-20 collision, jemalloc not actually linked, make-clean missing)
- A6: Install-mechanism side-by-side bluffs (phantom directory path, `alias tmux='tmx'` shadowing system `tmux`, stale ATMOSphere branding)
- A8: macOS bridge + side-by-side coexistence (later superseded by native dual-OS in this release)
- A10: Status-bar colour silently defaulted to green; bridge ignored macOS hostname
- A11: Triple-layer regression protection so A10 cannot re-occur
- A12: Constitution §11.4.7 — operator-path test coverage rule (every gate test MUST exercise the same entry point an end-user invokes)
- A13: Per-session cgroup isolation (each `tmx new -s X` in its own scope, OOM in one doesn't kill others)
- A14: This-release verification cycle — fresh runtime evidence captured for every claim

Install

macOS (Darwin Apple Silicon or Intel):

```
brew install podman # optional, for build_containerized.sh
git clone --recurse-submodules git@github.com:vasic-digital/
tmux.git ~/Projects/tmux
cd ~/Projects/tmux
bash scripts/setup.sh
```

Linux:

```
git clone --recurse-submodules git@github.com:vasic-digital/
tmux.git ~/Projects/tmux
cd ~/Projects/tmux
sudo bash scripts/install_deps.sh
bash scripts/setup.sh
```

After `setup.sh` reports GREEN: open a new shell (`source ~/.zshrc` or `source ~/.bashrc`). Then `tmx new -s anything` drops you into a session as your host user with the full host environment and kernel-enforced resource caps.

Usage

```
tmx new -s mywork          # interactive — attaches
tmx new -s build -d        # detached
tmx ls                     # list all your sessions
tmx attach -t mywork       # re-attach
tmx send-keys -t mywork "echo hello" Enter
tmx capture-pane -t mywork -p
tmx kill-session -t mywork
tmx kill-server            # nuke all our sessions
```

Per-session resource overrides:

```
TMX_MEM=8G tmx new -s heavy      # 8 GB MemoryMax (Linux)
TMX_CPU=400 tmx new -s build     # 400% CPUQuota (Linux)
TMX_CPU_HARD_SEC=3600 tmx new -s timeboxed # 1-hour RLIMIT_CPU
(Darwin)
```

Verification commands

Operators can confirm anti-bluff covenant compliance at any time:

```
bash scripts/setup.sh --verify-only # full 14-test gate →
expect GREEN
bash scripts/test_e2e.sh            # end-to-end → expect
PASS=9/0/0
TMX_TEST_DESTRUCTIVE=1 bash scripts/test_vm.sh # Linux
destructive suite
META=1 bash scripts/test_vm.sh      # paired-mutation harness
```

Coexistence with system tmux

tmx and the system tmux (Homebrew on macOS, distro package on Linux) coexist side-by-side. The bashrc/zshrc snippet PATH-prepends scripts/ so tmx is the project's wrapper; tmux resolves to whatever was on PATH before. No alias shadowing.

Submodules

- tmux/ — upstream tmux/tmux pinned to tag 3.6a (do not modify).
- Containers/ — vasic-digital/Containers Go module providing generic container orchestration primitives. Anti-bluff covenant
 - §11.4.7 propagated. Tag: b077f2c at release time.

Known limitations

- macOS memory cap is per-process via launchd-bsd-style limits (informational only — Darwin doesn't enforce RLIMIT_AS). For true per-session memory containment, run on Linux.
- The destructive test suite (tests 12 / 13 / 14) requires TMX_TEST_DESTRUCTIVE=1 and is Linux-only; on Darwin these SKIP per topology dispatch.

Acknowledgements

Built on the shoulders of:

- `tmux` / `tmux` upstream (3.6a tag)
- `jemalloc` (linked at build time)
- `libevent` + `ncurses` + `utf8proc` (Mach-O), `libtinfo` + `libevent_core` (ELF)
- `systemd-run` + `cgroup-v2` (Linux isolation primitive)
- Homebrew (macOS dependency provider)

Released under Apache 2.0 (see `LICENSE`).

[v1.0.1] — Unreleased

Post-release development cycle following the v1.0.0 cut.

- `VERSION` bumped to `1.0.1` / `versionCode=2` to satisfy the operator's strictly-increasing-version-code mandate immediately after a release.
- `released=` intentionally blank until the next tag is cut (gate value for "not-yet-released"; CI / package builders MUST refuse to publish unless `released=` is populated with the cut date).
- No functional code changes in this entry — this is the post-release bump itself. Subsequent fixes, refactors, and new features will append bullets above this paragraph.